

Normalizing flow f_0 in one slide

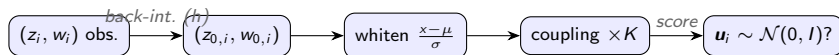
Idea: model the unknown initial DF $f_0(z_0, w_0)$ as a *simple source density deformed by a trainable bijection*.

- **Source density:** 2-D standard Gaussian, $\mathbf{u} \sim \mathcal{N}(\mathbf{0}, I)$, $p_{\text{base}}(\mathbf{u}) = \frac{1}{2\pi} e^{-\|\mathbf{u}\|^2/2}$.
- **Bijection:** stack of K RealNVP *affine coupling layers* — keep one coordinate, affinely move the other, with scale/shift a small neural net of the kept one (roles alternate layer to layer):

$$(a, b) \mapsto \left(a, (b - t(a)) e^{-s(a)} \right), \quad s, t = \text{MLP}(a).$$

Invertible in closed form: a is untouched, so $s(a), t(a)$ are recomputable — the MLP is never inverted.

- **Jacobian is free:** triangular $\Rightarrow \log|\det J_k| = -s_k(a)$; over the stack they add.
- **Density** (change of variables, physical units):
 $\log f_0^{\text{flow}}(z_0, w_0) = -\frac{1}{2} \|\mathbf{u}\|^2 - \log 2\pi + \sum_k (-s_k) - \log(\sigma_z \sigma_w).$



Static Gaussian vs. normalizing flow: the likelihoods

Both back-integrate each star through the trial potential, $(z_i, w_i) \xrightarrow{h} (z_{0,i}, w_{0,i})$, and use Liouville ($f_{\text{obs}} = f_0$ along orbits; symplectic $\Rightarrow$ no Jacobian). **Only the role of f_0 differs.**

Static Gaussian f_0

$$\ln \mathcal{L}(h) = - \sum_i \left[\frac{z_{0,i}(h)^2}{2\bar{\sigma}_z^2} + \frac{w_{0,i}(h)^2}{2\bar{\sigma}_w^2} \right] + c,$$

$$\max_{h \text{ only}} \ln \mathcal{L}(h)$$

- $\bar{\sigma}_z, \bar{\sigma}_w$ **frozen** (possibly wrong).
- Fixed target shape $\Rightarrow h$ absorbs any f_0 mismatch $\Rightarrow$ **bias**, but sharp (± 1 pc, init-independent).

Flow f_0 (Test 9/10)

$$\ln \mathcal{L}(h, \theta) = \sum_i \ln f_\theta(z_{0,i}(h), w_{0,i}(h)),$$

$$\max_{h, \theta \text{ jointly}} \ln \mathcal{L}(h, \theta)$$

- θ = coupling nets: target **reshapes itself** to the cloud of the current $h \Rightarrow$ no mismatch to absorb $\Rightarrow$ **no parametric bias**.
- But f_θ can fit the cloud of *any* $h \Rightarrow$ likelihood flattens; h pinned only by *finite capacity* (flow can't fit wound-up spirals of wrong h).

Same data, physics, estimator — one line changes: $\mathcal{N}_{\bar{\sigma}} \rightarrow f_\theta$, with θ joining the fit. Rigid f_0 : sharp but biased. Flow f_0 : unbiased but only capacity-constrained.